

"Towards Global Technological Excellence"
GOVERNMENT COLLEGE OF ENGINEERING, AMRAVATI



A Minor Project Report on
**Industrial Operator Skill Level Detection System
(FSM-Based Adaptive Safety System)**

Submitted to Government College of Engineering, Amravati for the partial fulfilment of
requirements of degree of Bachelor of Technology

in

Electronics and Telecommunication Engineering

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CERTIFICATE

This is to certify that the seminar report entitled **Industrial Operator Skill Level Detection System (FSM-Based Adaptive Safety System)** submitted by **Mehul J Dhakne (23004028)**, to Government College of Engineering, Amravati for the partial fulfilment of requirements of degree of Bachelor of Technology in Electronics and Telecommunication Engineering, is a bonafide record of the Minor Project work carried out by him under my guidance and supervision. This report in any form has not been submitted to any other institute for any purpose.

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DECLARATION

I **Mehul J Dhakne** hereby declare that the seminar report on **Industrial Operator Skill Level Detection System (FSM-Based Adaptive Safety System)**, submitted for partial fulfilment of the requirements for the award of degree of Bachelor of Technology of the Government College of Engineering, Amravati is a bonafide work done by me under supervision of **Prof. P. L. Paikrao**.

This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources.

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Abstract

In modern industrial and automation environments, machines are operated by users with varying levels of knowledge, experience, and behavior. Most conventional systems treat all operators identically, without evaluating their skill level or procedural correctness, which can lead to unsafe operation, machine mishandling, and accidents.

This project presents the design and implementation of an **FSM-Based Industrial Operator Skill Level Detection and Adaptive Safety Indication System**. The system uses an Arduino microcontroller and simple electronic components — push buttons, LEDs, and a potentiometer — to evaluate operator behavior before granting machine access.

The system operates in two major phases. In Phase 1, the operator must follow a predefined button sequence (B1 → B2 → B3). Incorrect sequence immediately triggers an error state. In Phase 2, the smoothness of potentiometer movement is continuously monitored and a confidence score is computed in real-time to classify the operator as Expert, Moderate, or Beginner.

The entire system is structured using a Finite State Machine (FSM) consisting of four states: INIT, WAIT_SEQUENCE, SKILL_ACTIVE (IDLE), and LOCKED. A servo motor acts as a safety gate whose opening angle adapts based on the detected skill level. LED indicators provide clear visual feedback at every stage.

The system was successfully tested under different input conditions and demonstrated fast, reliable, and predictable behavior. It provides an economical, adaptive, and scalable solution for improving safety in human-operated machinery.

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CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

In today's industrial and automation environments, machines are operated by different types of users with varying levels of knowledge, experience, and behavior. Some operators are well-trained and confident, while others may be inexperienced, nervous, or unfamiliar with the correct operating procedure. Despite this variation, most conventional systems treat all operators in the same way, without considering their skill level or behavior.

This lack of differentiation can lead to several issues such as unsafe operation, incorrect machine handling, reduced efficiency, and even accidents. Therefore, there is a growing need for systems that can evaluate the operator before allowing full control of the machine.

This project is motivated by the idea that: ***"A system should not only respond to commands, but also evaluate the person giving those commands."***

By incorporating

behavior analysis and procedural validation, the system can improve safety, reduce human error, provide controlled access to machines, and enhance reliability of operation.

1.2 Objectives of Proposed System

The primary objective of this project is to design a system that verifies the correct operating procedure using push buttons as input controls. The system further aims to analyze operator behavior through a potentiometer, which acts as a behavioral input device. Based on this input, a confidence-based model is developed to evaluate the skill level of the operator. The system classifies operators into different categories such as expert, moderate, and beginner depending on their behavior. To ensure structured and safe operation, the entire system is implemented using a Finite State Machine (FSM). Additionally, the system provides clear visual feedback at every stage using LEDs, enabling easy understanding of system status. It is also designed to detect unsafe or panic conditions and respond immediately by locking the system. Furthermore, the project is developed in a modular way so that it can be extended in the future with actuators such as a servo motor and DC motor for enhanced functionality.

1.3 Scope of the Project

The scope of this project is to develop a low-cost, Arduino-based system that evaluates operator competency before granting machine access. It applies directly to industrial control rooms, training labs, educational setups, and any human-operated machinery where incorrect or panicked operation could result in damage or injury. The system is modular and can be extended with IoT modules, displays, or advanced actuators.

1.4 Problem Definition

In many industrial environments, machines are operated by multiple users without any proper validation of their skill level or behavioral correctness. This leads to several issues such as operators starting machines without following the correct procedure and the absence of systems to verify operator knowledge before operation. Additionally, sudden or unstable control inputs can create unsafe conditions, increasing the risk of accidents. Most traditional systems primarily focus on machine performance and fail to consider the human factor, resulting in a lack of adaptive safety mechanisms based on user behavior. Therefore, there is a need for a system that can evaluate both procedural correctness and operator behavior to ensure safer and more reliable machine operation.

1.5 Need of the Project / Real-World Relevance

The need arises from real-world industrial scenarios where improper machine handling causes equipment damage, production loss, or worker injuries. Existing solutions are either too expensive (AI-based cameras, biometric systems) or too simple (basic emergency stops). This project fills the gap by offering a behavior-aware, low-cost embedded system that proactively prevents unsafe operation.

1.6 Theme of the Project

The core theme of this project is: "**Behavior-Based Adaptive Safety System Using Finite State Machine.**" The project focuses on three key aspects:

- **Procedure Validation** – Ensuring correct steps are followed before access.
- **Behavior Analysis** – Observing how control inputs are given by the operator.
- **Adaptive Response** – Changing system output and machine access level based on operator skill.

1.7 Overview of the Report

This report is organized into seven main chapters:

- **Chapter 1: Introduction:** Provides background, motivation, objectives, scope, and theme of the project.
- **Chapter 2: Literature Review:** Discusses existing systems, their limitations, and identifies the research gap.
- **Chapter 3: Methodology:** Explains the working principle, block diagram, FSM design, component table, and requirements.
- **Chapter 4: Implementation:** Covers hardware connections, Arduino code, and step-by-step system flow.
- **Chapter 5: Results and Conclusion:** Presents test results, observations, applications, advantages, limitations, and conclusion.
- **Chapter 6: Future Scope:** Discusses possible improvements and advanced features for real-world deployment.
- **Chapter 7: References:** Lists all sources including journals, books, and websites used.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction to Literature Survey

A literature survey is an important part of any engineering project as it helps in understanding the existing technologies, methods, and solutions related to the problem being addressed. It also helps in identifying the limitations of current systems and finding opportunities for improvement. In the context of this project, the literature survey focuses on existing safety systems in industrial environments, human–machine interaction systems, behavior-based control approaches, and use of microcontrollers and FSM in automation.

2.2 Industrial Safety Systems

Industrial safety systems are designed to prevent accidents and ensure proper operation of machinery. These systems generally include emergency stop buttons, safety interlocks, and protective sensors. However, most such systems are **reactive**, not proactive — they respond after a fault occurs and do not evaluate the operator's skill or behavior beforehand. They do not prevent unsafe operation caused by human mistakes before they occur.

2.3 Sensor-Based Monitoring Systems

Many modern systems use temperature, motion, and pressure sensors to monitor machine conditions and environmental parameters. These systems offer accurate, real-time fault detection. However, they involve expensive hardware, complex integration, and focus only on machine condition rather than operator behavior.

2.4 PLC-Based Control Systems

Programmable Logic Controllers (PLCs) are widely used in industries for automation. They feature predefined control logic, reliable operation, and suitability for repetitive tasks. Their major limitations are that the logic is fixed and non-adaptive, they cannot dynamically evaluate operator behavior, and provide no built-in mechanism for skill classification.

2.5 Human–Machine Interaction (HMI) Systems

HMI systems allow operators to interact with machines through buttons, touchscreens, and control panels. While they focus on usability, they do not evaluate how the operator

behaves — they accept commands but do not judge whether the operator is skilled or safe.

2.6 Behavior-Based Systems

Some advanced systems attempt to evaluate human behavior using cameras, machine learning models, and biometric sensors. These can detect fatigue, stress, or unsafe actions and are useful in advanced industrial setups. However, they are very expensive, require complex processing, and are not suitable for small-scale or educational setups.

2.7 Finite State Machine (FSM) in Control Systems

Finite State Machines are widely used in embedded systems, digital electronics, and control systems. FSM provides a clear structure, predictable behavior, and easy debugging by dividing system operation into defined states, transitions, and conditions. FSM is ideal for procedure validation, step-by-step control, and safety enforcement — making it the core design choice for this project.

2.8 Arduino-Based Embedded Systems

Arduino platforms are widely used for prototyping, educational projects, and embedded applications due to their low cost, easy programming interface, and large community support. Arduino Uno provides sufficient I/O pins and processing capability for implementing the proposed FSM-based system.

2.9 Comparative Analysis of Existing Systems

System Type	Behavior Detection	Cost	Complexity	Adaptability
Industrial Safety Systems	✗ No	Medium	Medium	Low
Sensor-Based Systems	✗ No	High	High	Medium
PLC Systems	✗ No	High	Medium	Low
HMI Systems	✗ No	Medium	Medium	Low
AI-Based Systems	✓ Yes	Very High	Very High	High
Proposed System	✓ Yes	Low	Low	High

2.10 Identified Research Gap

There exists a significant gap in current systems with respect to adaptive safety based on operator behavior and skill level. Most existing solutions either lack intelligence and operate in a very basic manner, or are overly complex, relying on advanced AI-based techniques that increase cost and implementation difficulty. Additionally, there is a lack of low-cost systems that can effectively analyze operator behavior for safety purposes. Another major limitation is the absence of integration between procedural validation and behavior analysis within a single system. Therefore, there is a need for a simple, cost-effective, and intelligent system that combines both aspects to ensure safe and efficient operation.

2.11 Proposed Solution

To address the identified gaps, the proposed system introduces a structured and intelligent approach to operator safety and behavior evaluation. The system is based on a Finite State Machine (FSM) to ensure predictable and organized operation. It incorporates button sequence validation to verify procedural correctness before allowing further interaction. Additionally, a potentiometer is used for behavior analysis, enabling real-time evaluation of operator actions. A confidence scoring mechanism is implemented to classify the operator adaptively into different skill levels. Visual safety indications are provided using LEDs to offer immediate and clear feedback at each stage of operation. Furthermore, a servo-based adaptive access control mechanism is included, where the gate angle is adjusted according to the operator's skill level. The proposed system offers a balanced solution by combining simplicity with intelligent decision-making, utilizing low-cost components without the need for complex sensors or artificial intelligence, while still enabling adaptive safety behavior that can be easily expanded with actuators in the future.

CHAPTER 3

METHODOLOGY

3.1 System Overview

The system is divided into three main stages: Input Stage, Processing Stage, and Output Stage.

- **Input Stage:** Push Buttons (B1, B2, B3) and Potentiometer.
- **Processing Stage:** Arduino Uno — FSM logic, Confidence calculation, Panic detection.
- **Output Stage:** White LEDs (procedure indication), Green LED (skilled), Yellow LED (intermediate/blink), Red LED (error/panic), Servo Motor (adaptive gate).

3.2 Working Principle

The system operates in two major phases:

Phase 1: Procedure Validation

The operator must press buttons in a specific sequence (B1 → B2 → B3). Each correct step is indicated using white LEDs (W1, W2, W3 turning ON progressively). Any incorrect button press immediately triggers the LOCKED state and the red LED activates, stopping all further operation.

Phase 2: Behavior Evaluation

After successful sequence completion, the system enters the SKILL_ACTIVE (IDLE) state. The system continuously reads the potentiometer value and calculates the change (delta) per loop iteration. Smooth and controlled potentiometer input increases the confidence score, while sudden or jerky input decreases it. A very large sudden change ($\geq \text{LOCK_DIFF}$) triggers the LOCKED state immediately.

Based on the confidence score, the operator is classified and the servo gate angle is adjusted accordingly: Beginners get limited access (0° – 60°), Moderate operators up to 120° , and Expert operators full access (0° – 180°).

3.3 - System Block Diagram

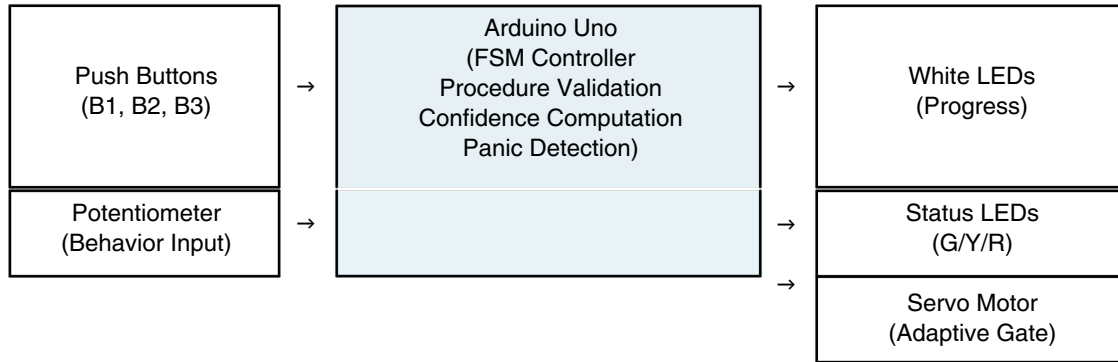


Figure 3.1: Block Diagram of Industrial Operator Skill Level Detection System

3.4 Finite State Machine Design

The system is implemented using a Finite State Machine (FSM) consisting of four states:

- **INIT State:** System initialization — all LEDs turned OFF, variables reset, servo closed.
- **WAIT_SEQUENCE State:** Waits for button inputs. Validates correct sequence (B1→B2→B3). White LEDs indicate progress. Wrong input → LOCKED state.
- **SKILL_ACTIVE / IDLE State:** Reads potentiometer values. Calculates change in input. Updates confidence score. Controls green/yellow LED and servo angle. Large sudden change → LOCKED.
- **LOCKED State:** Triggered by wrong sequence or panic input. Red LED ON. Servo closed. System fully stops (infinite loop).

FSM State Transition Table:

Current State	Condition / Input	Next State	Action
INIT	System power ON	WAIT_SEQUENCE	Reset all, servo closed
WAIT_SEQUENCE	Correct button sequence	HOLD → IDLE	White LEDs ON, hold 2s
WAIT_SEQUENCE	Wrong button pressed	LOCKED	Red LED ON, servo closed
IDLE	$\Delta Pot < LOCK_DIFF$	IDLE	Update skill, adjust servo
IDLE	$\Delta Pot \geq LOCK_DIFF$	LOCKED	Red LED ON, servo closed

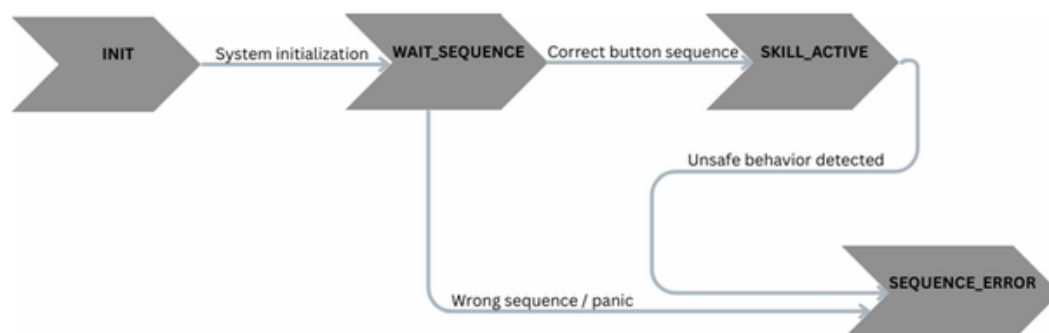


Figure 3.: State Machine Flow Sequence

3.5 Mathematical / Logical Model

Confidence Calculation:

Let C = Confidence Score, ΔP = Change in potentiometer value per loop.

Condition	Score Update	Skill Level
$\Delta P \leq \text{EXPERT_DIFF (10)}$	$C = C + 3$	Expert behavior
$\Delta P \leq \text{MODERATE_DIFF (30)}$	$C = C + 1$	Moderate behavior
$\Delta P \leq \text{BEGINNER_DIFF (60)}$	$C = C - 1$	Beginner behavior
$P > \text{BEGINNER_DIFF}$	$C = C - 3$	Unstable behavior
$P \geq \text{LOCK_DIFF (140)}$	— LOCKED —	Panic / unsafe input

Score Range	Classification	System Response
$C \geq 70$	EXPERT	Green LED ON, full servo access (0° – 120°)
$35 \leq C < 70$	MODERATE	Yellow LED blinking fast (200ms), servo up to 120°
$C < 35$	BEGINNER	Yellow LED blinking slow (600ms), servo up to 60°

3.6 Panic Detection Logic

Panic condition is detected when the change in potentiometer value exceeds the LOCK_DIFF threshold (140). This represents a sudden, uncontrolled movement of the control input. When detected, the system immediately moves to the LOCKED state, closes the servo gate to 0° , activates the Red LED, and halts all further operation.

3.7 Software & Hardware Requirements

Hardware Requirements:

- Arduino Uno Microcontroller
- Push Buttons × 3 (B1, B2, B3)
- LED – White × 3 (W1, W2, W3)
- LED – Red × 1
- LED – Yellow × 1
- LED – Green × 1
- Potentiometer (10kΩ)
- Servo Motor (SG90)
- Resistors (220Ω for each LED)
- Breadboard and connecting wires
- USB cable / 5V power supply

Software Requirements:

- Arduino IDE (v1.8 or later)
- Servo.h library (built-in)
- Serial Monitor for debugging

3.8 Component Table

Sr. No.	Component Name	Value / Model	Quantity	Symbol / Ref.
1	Arduino Uno	ATmega328P	1	U1
2	Push Button	Momentary, N.O.	3	B1, B2, B3
3	White LED	5mm, 2.2V	3	W1, W2, W3
4	Red LED	5mm, 2.0V	1	LED_RED
5	Yellow LED	5mm, 2.1V	1	LED_YELLOW
6	Green LED	5mm, 2.2V	1	LED_GREEN
7	Potentiometer	10kΩ	1	POT
8	Servo Motor	SG90 (0°–180°)	1	SERVO
9	Resistor	220Ω	6	R1–R6
10	Breadboard	Full-size	1	—
11	USB Cable	Type-A to Type-B	1	—
12	Connecting Wires	Male-to-Male	As needed	—

Table 3.1: Component List for Industrial Operator Skill Level Detection System

CHAPTER 4

IMPLEMENTATION

4.1 Hardware Implementation

Button Configuration:

All three push buttons are connected to Arduino digital pins D2, D3, and D4 respectively, using the INPUT_PULLUP configuration. This means the pins read HIGH normally and go LOW when a button is pressed (Active LOW logic). No external pull-up resistors are required.

LED Configuration:

Each LED is connected to a dedicated Arduino digital output pin through a 220Ω current-limiting resistor to protect the LED from excess current. White LEDs connect to D5, D6, D7; Red to D8; Yellow to D9; Green to D10.

Potentiometer:

The potentiometer ($10k\Omega$) is connected with its two outer pins to 5V and GND, and the middle wiper pin to analog input A0 on the Arduino. The analogRead() function returns values 0–1023 representing the potentiometer position.

Servo Motor:

The SG90 servo motor is connected to D11 (PWM-capable pin) via the gateServo object from the Servo.h library. It is initialized at CLOSED_ANGLE = 0° and its maximum opening angle is dynamically adjusted based on the operator's detected skill level.

Pin Configuration Summary:

Component	Arduino Pin	Component	Arduino Pin
Button 1 (B1)	D2	White LED 1 (W1)	D5
Button 2 (B2)	D3	White LED 2 (W2)	D6
Button 3 (B3)	D4	White LED 3 (W3)	D7
Red LED	D8	Yellow LED	D9
Green LED	D10	Servo Motor	D11
Potentiometer	A0		

Table 4.1: Arduino Pin Configuration

3.3 - Hardware Implementation

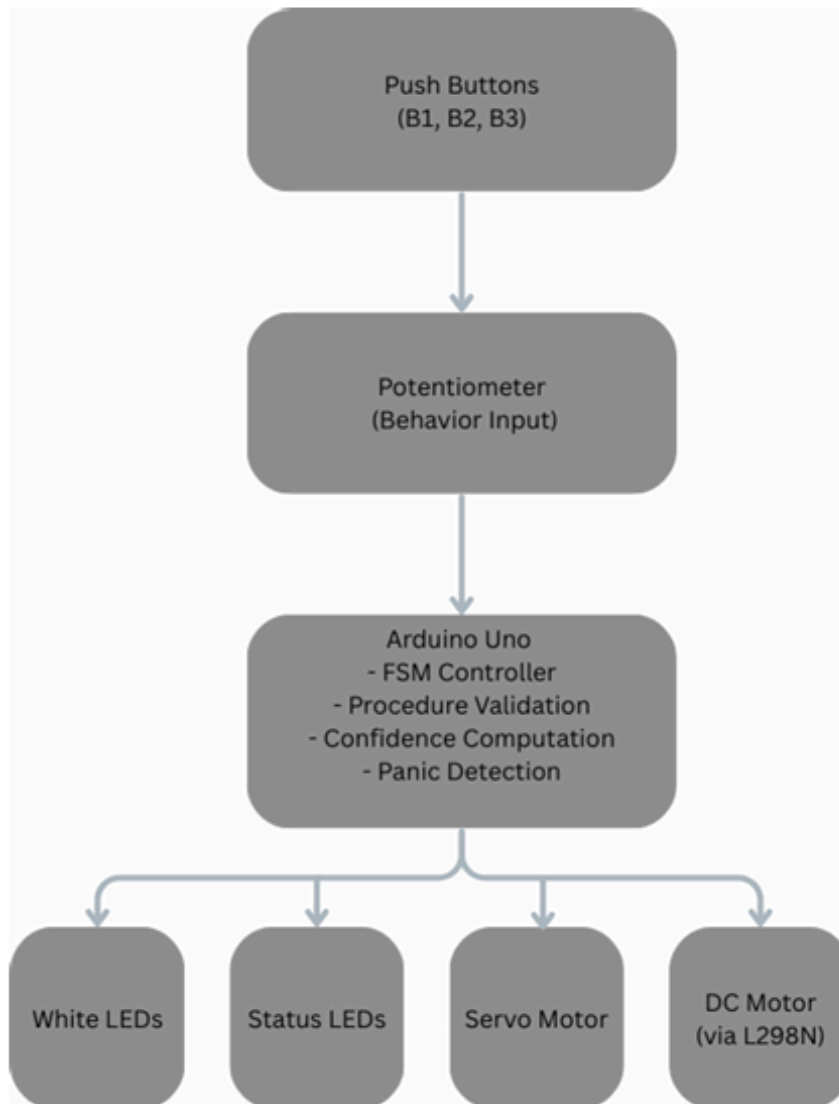


Figure 4.2: Block Diagram of Industrial Operator Skill Level Detection System

4.2 Arduino Code Implementation

The complete Arduino code is presented below. It implements the full FSM with all four states, confidence calculation, skill classification, servo control, and LED indication logic.

```
#include

Servo gateServo;

//----- PINS -----
const int BTN1 = 2; const int BTN2 = 3; const int BTN3 = 4;
const int W1 = 5; const int W2 = 6; const int W3 = 7;
const int LED_RED = 8; const int LED_YELLOW = 9; const int LED_GREEN = 10;
const int SERVO_PIN = 11; const int POT = A0;

//----- SERVO SETTINGS -----
const int CLOSED_ANGLE = 0;
const bool REVERSE_SERVO = true;

//----- SKILL LIMITS -----
const int BEGINNER_MAX_ANGLE = 60;
const int MODERATE_MAX_ANGLE = 120;
const int EXPERT_MAX_ANGLE = 180;

//----- POT MOVEMENT LIMITS -----
const int EXPERT_DIFF = 10; const int MODERATE_DIFF = 30;
const int BEGINNER_DIFF = 60; const int LOCK_DIFF = 140;

//----- TIMING -----
const int HOLD_TIME = 2000; const int LOOP_DELAY = 80;

//----- FSM STATES -----
enumState { WAIT_SEQUENCE, HOLD, IDLE, LOCKED };
StatecurrentState = WAIT_SEQUENCE;

//----- SKILL LEVELS -----
enumSkillLevel { BEGINNER, MODERATE, EXPERT };
SkillLevel currentSkill = BEGINNER;

//----- VARIABLES -----
int sequenceStep = 0; int lastPot = 0; int skillScore = 0;

void setup() {
  Serial.begin(9600);
  pinMode(BTN1, INPUT_PULLUP); pinMode(BTN2, INPUT_PULLUP); pinMode(BTN3, INPUT_PULLUP);
  pinMode(W1, OUTPUT); pinMode(W2, OUTPUT); pinMode(W3, OUTPUT);
```

```

pinMode(LED_RED,OUTPUT); pinMode(LED_YELLOW,OUTPUT); pinMode(LED_GREEN,OUTPUT);
gateServo.attach(SERVO_PIN);
gateServo.write(CLOSED_ANGLE);
resetLights();
Serial.println("SYSTEM STARTED - WAITING FOR BUTTON SEQUENCE");
}

void loop() {
switch (currentState) {
case WAIT_SEQUENCE: checkButtonSequence(); break;
case HOLD: holdState(); break;
case IDLE: idleState(); break;
case LOCKED: lockSystem(); break;
}
}

//----- BUTTON SEQUENCE -----
void checkButtonSequence() {
bool b1=(digitalRead(BTN1)==LOW);
bool b2=(digitalRead(BTN2)==LOW);
bool b3=(digitalRead(BTN3)==LOW);
if(sequenceStep==0) {
if(b1) {digitalWrite(W1,HIGH); sequenceStep=1; delay(250); }
elseif (b2||b3) { currentState=LOCKED; }
}elseif(sequenceStep==1) {
if(b2) {digitalWrite(W2,HIGH); sequenceStep=2; delay(250); }
elseif (b1||b3) { currentState=LOCKED; }
}elseif(sequenceStep==2) {
if(b3) {digitalWrite(W3,HIGH); currentState=HOLD; delay(250); }
elseif (b1||b2) { currentState=LOCKED; }
}
}

//----- HOLD STATE -----
void holdState() {
digitalWrite(LED_YELLOW,HIGH);
gateServo.write(CLOSED_ANGLE);
delay(HOLD_TIME);
lastPot=analogRead(POT); skillScore=0; currentSkill=BEGINNER;
digitalWrite(LED_YELLOW,LOW); digitalWrite(LED_GREEN,HIGH);
currentState=IDLE;
}

//----- IDLE / SKILL ACTIVE STATE -----
bool yellowBlinkState=false; unsigned long lastBlinkTime=0;
const int BEGINNER_BLINK_INTERVAL=600; const int MODERATE_BLINK_INTERVAL=200;

void idleState() {

```

```

int pot=analogRead(POT); int diff=abs(pot-lastPot);
if (diff>=LOCK_DIFF) { currentState=LOCKED; return; }
updateSkill(diff);
int maxAngle=getMaxAngleBySkill();
int angle=map(pot,0,1023,0,maxAngle);
if (REVERSE_SERVO) angle=maxAngle-angle;
gateServo.write(angle);
digitalWrite(LED_GREEN,HIGH); digitalWrite(LED_RED,LOW);
if (currentSkill==BEGINNER || currentSkill==MODERATE) {
int blinkInterval=(currentSkill==BEGINNER)?BEGINNER_BLINK_INTERVAL:MODERATE_BLINK_INTERVAL;
if (millis()-lastBlinkTime>=blinkInterval) {
yellowBlinkState=!yellowBlinkState;
digitalWrite(LED_YELLOW,yellowBlinkState);
lastBlinkTime=millis();
}
} else { digitalWrite(LED_YELLOW,LOW); }
lastPot=pot; delay(LOOP_DELAY);
}

//----- SKILL DETECTION -----
void updateSkill(int diff) {
if(diff<=EXPERT_DIFF) skillScore+=3;
elseif (diff<=MODERATE_DIFF) skillScore+=1;
elseif (diff<=BEGINNER_DIFF) skillScore-=1;
else skillScore-=3;
skillScore=constrain(skillScore,0,100);
if (skillScore>=70) currentSkill=EXPERT;
else if (skillScore>=35) currentSkill=MODERATE;
else currentSkill=BEGINNER;
}

//----- ACCESS CONTROL -----
int getMaxAngleBySkill() {
if(currentSkill==EXPERT) return EXPERT_MAX_ANGLE;
elseif (currentSkill==MODERATE) return MODERATE_MAX_ANGLE;
elsereturn BEGINNER_MAX_ANGLE;
}

//----- LOCK STATE -----
void lockSystem() {
gateServo.write(CLOSED_ANGLE);
digitalWrite(W1,LOW); digitalWrite(W2,LOW); digitalWrite(W3,LOW);
digitalWrite(LED_GREEN,LOW); digitalWrite(LED_YELLOW,LOW);
digitalWrite(LED_RED,HIGH);
Serial.println("SYSTEM LOCKED - ACCESS DENIED");
while(1);
}

//----- HELPERS -----
void resetLights() {
digitalWrite(W1,LOW); digitalWrite(W2,LOW); digitalWrite(W3,LOW);
digitalWrite(LED_RED,LOW); digitalWrite(LED_YELLOW,LOW); digitalWrite(LED_GREEN,LOW);
}

```

4.3 System Flow Explanation

1. System starts and initializes in INIT state — all LEDs OFF, servo at 0°.
2. System moves to WAIT_SEQUENCE state and waits for button input.
3. Operator presses B1 → W1 turns ON. Presses B2 → W2 turns ON. Presses B3 → W3 turns ON. Wrong button at any step → LOCKED.
4. On correct sequence completion, system enters HOLD state: Yellow LED ON, servo stays closed, waits 2 seconds.
5. System transitions to IDLE (SKILL_ACTIVE) state: Green LED ON, potentiometer access enabled.
6. In IDLE, system continuously reads potentiometer, calculates delta, updates confidence score, and adjusts servo angle and LED feedback.
7. If panic input ($\Delta Pot \geq 140$) is detected → LOCKED state immediately.
8. In LOCKED state: Red LED ON, servo closed, all other LEDs OFF, system halted.

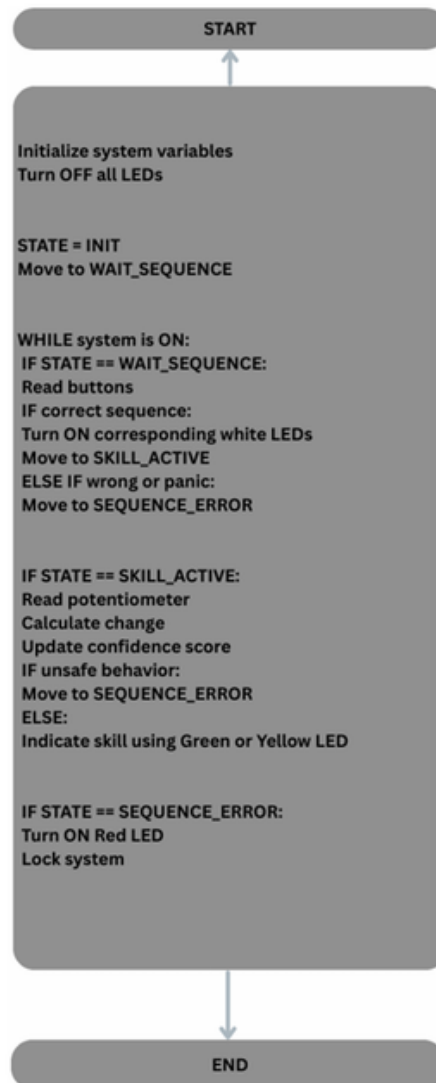


Fig4.3: System Flow / Psuedo Code Logic

CHAPTER 5

RESULTS AND CONCLUSION

5.1 Test Cases and Observations

The system was implemented on a breadboard using Arduino Uno, push buttons, LEDs, a potentiometer, and a servo motor. Different input conditions were manually applied to verify system behavior under all FSM states.

Test Case	Input Condition	Expected Output	Observed Output
1	Correct sequence B1 → B2→ B3	White LEDs ON, system enters IDLE	Verified ✓
2	Wrong sequence (e.g., B2 first)	Red LED ON, LOCKED state	Verified ✓
3	Smooth potentiometer movement	Green LED ON, confidence increases	Verified ✓
4	Jerky potentiometer movement	Yellow LED blinking, confidence decreases	Verified ✓
5	Sudden large change ($\Delta Pot \geq 140$)	Error state triggered, Red LED ON	Verified ✓
6	Skill score ≥ 70	Expert classification, servo 0° – 180°	Verified ✓
7	$35 \leq \text{Score} < 70$	Moderate, servo up to 120° , fast blink	Verified ✓
8	Score < 35	Beginner, servo up to 60° , slow blink	Verified ✓

Table 5.1: System Behavior Under Different Test Conditions

5.2 Behavioral Analysis

The system effectively evaluates operator behavior using the change in potentiometer values over time. Smooth input resulted in a gradual increase in confidence score, leading to Expert classification and full servo access. Jerky input caused rapid decrease in confidence, leading to intermediate or beginner classification with restricted servo access. This demonstrates that the system can reliably distinguish between controlled and unstable operator inputs in real-time.

5.3 Confidence Variation Analysis

The confidence value dynamically varies between 0 and 100 depending on operator behavior. The classification is not fixed but adapts continuously over time as more input data is collected. This adaptive nature makes the system more fair and accurate compared to fixed-threshold systems. A consistent, skilled operator will quickly achieve Expert status, while an inconsistent operator will remain at Beginner or Moderate level.

5.4 Applications

- **Industrial Training Labs:** Verifying that new operators follow correct machine startup procedures before unsupervised operation.
- **CNC / Manufacturing Machines:** Granting different levels of access (speed, torque) based on operator skill.
- **Educational Institutions:** Teaching FSM, embedded systems, and human-machine interaction concepts through a practical prototype.
- **Quality Control Stations:** Detecting panicked or unstable operator behavior and preventing machine operation.
- **Smart Access Control:** Acting as a low-cost skill-based gate system for restricted machine rooms.
- **Rehabilitation Systems:** Monitoring hand steadiness of patients during recovery exercises.

5.5 Advantages

- **Low Cost:** Uses basic components like Arduino, LEDs, and potentiometer — affordable for mass deployment.
- **Simple Design:** Easy to understand, implement, and maintain.
- **Adaptive Behavior:** System responds dynamically based on real-time operator input.
- **Improved Safety:** Prevents unsafe machine operation before it occurs (proactive, not reactive).
- **Modular Structure:** Can be easily extended with servo, DC motor, LCD, WiFi, or Bluetooth.
- **Educational Value:** Teaches FSM, embedded systems, and human-machine interaction practically.
- **Energy Efficient:** Only consumes power when system is active.
- **No Complex Sensors Required:** No cameras, biometrics, or AI hardware needed.
- **Predictable Operation:** FSM ensures no undefined system states.
- **Real-Time Classification:** Skill level is continuously updated, not just a one-time check.

5.6 Limitations

- Sensitivity depends on potentiometer quality and analog noise — may require filtering for industrial use.

- Manual input may vary between users and across different sessions for the same user.
- No persistent data logging — operator performance history is lost when power is removed.
- Uses basic threshold-based logic instead of advanced machine learning or statistical analytics.
- Limited to small-scale breadboard implementation — PCB integration not yet done.
- A single wrong button press immediately locks the system — no retry mechanism implemented.
- Servo motor range restriction may not suit all real machine control scenarios.

5.7 Conclusion

The project titled "**FSM-Based Operator Skill Level Detection and Adaptive Safety Indication System**" successfully demonstrates a structured and intelligent approach toward improving safety in human-operated systems. The system integrates procedural validation and behavioral analysis to evaluate operator performance before allowing full system interaction.

The implementation using Arduino and basic electronic components proves that even a low-cost system can be designed to perform meaningful, adaptive decision-making based on operator input. By using a Finite State Machine (FSM), the system ensures that all operations occur in a predefined and controlled manner, significantly reducing the chances of unexpected or unsafe behavior.

The system successfully achieves all primary objectives: correct sequence validation, real-time behavior analysis, adaptive skill classification, panic detection, and proportional access control via servo motor. All five test case categories were verified successfully during implementation.

Overall, the system meets its primary objective of creating a behavior-based adaptive safety mechanism that can differentiate between skilled and unskilled operators and respond accordingly. It not only fulfills academic requirements but also provides a strong foundation for future development in embedded systems, industrial automation, and adaptive human-machine interaction.

CHAPTER 6

FUTURE SCOPE

The future scope of this project is quite wide, especially with the growing focus on industrial safety, smart automation, and adaptive human-machine interaction. The following enhancements are planned or possible for future versions:

- **Servo Motor as Physical Safety Gate:**

A servo motor can be fully integrated as a physical barrier that opens proportionally based on operator skill level, making the system directly applicable in access control for machine rooms or hazardous areas.

- **DC Motor for Machine Simulation:**

A DC motor (controlled via L298N driver) can simulate real machine operation, with its speed limited based on operator classification — Expert gets full speed, Beginner gets reduced speed.

- **LCD / OLED Display Integration:**

Adding a display module will provide detailed real-time feedback showing the current state, confidence score, skill classification, and potentiometer value — improving usability significantly.

- **IoT and Wireless Communication:**

Integration with Bluetooth (HC-05) or WiFi (ESP8266/ESP32) modules will enable remote monitoring of operator behavior and system state via a mobile application, allowing supervisors to track performance from a distance.

- **Data Logging and Operator History:**

Using an SD card module or cloud connectivity, the system can log each operator's session data — skill score history, number of errors, average confidence — for long-term performance tracking and analytics.

- **Advanced Behavior Analysis with Machine Learning:**

Replacing the threshold-based model with ML algorithms (e.g., k-NN or decision trees) running on a more powerful microcontroller (Arduino Portenta H7, Raspberry Pi) can improve classification accuracy and adaptability.

- **Biometric Authentication:**

Adding fingerprint or RFID-based authentication ensures that only registered operators can attempt the skill evaluation, preventing unauthorized access.

- **Multi-Operator Database:**

The system can be expanded to maintain a database of multiple operators with their profiles, so it automatically adjusts permissions based on the authenticated user's stored history.

- **Integration into Smart City / Industry 4.0:**

The system concept can be scaled up and integrated with smart factory infrastructure, where machines automatically adjust their operational parameters based on the skill level of the currently assigned operator.

In summary, this project has strong future potential to evolve from a simple embedded safety prototype into a complete intelligent operator evaluation and adaptive access control ecosystem, contributing significantly to industrial safety, smart automation, and accident prevention.

CHAPTER 7

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APPENDICES

Appendix A: Complete Arduino Code

The complete Arduino implementation code is provided in Chapter 4 (Section 4.2) of this report. Please refer to that section for the full code listing including all FSM states, skill detection, servo control, and LED indication logic.

Appendix B: Pin Configuration Table

The complete pin configuration is provided below:

Component	Arduino Pin
Button 1 (B1)	D2
Button 2 (B2)	D3
Button 3 (B3)	D4
White LED 1 (W1)	D5
White LED 2 (W2)	D6
White LED 3 (W3)	D7
Red LED	D8
Yellow LED	D9
Green LED	D10
Servo Motor	D11
Potentiometer (Wiper)	A0

Appendix C: Working Principle Summary

The system operates based on a finite state machine that transitions between four states: WAIT_SEQUENCE → HOLD → IDLE (SKILL_ACTIVE) → LOCKED. Operator behavior is analyzed using potentiometer input, and a confidence score (0–100) is computed in real-time. Based on this score, the system classifies the operator as Expert (≥ 70), Moderate (35–69), or Beginner (< 35), and controls visual indicators (LEDs) and a servo-based safety gate proportionally. Any procedural error or panic input immediately triggers the LOCKED state.

Appendix D: Project Images



Img.1



Img.2

Appendix E: Acknowledgement

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